

Feature Documentation

1. Project Overview

Feature Name: Simple Escrow Payment with Digital Signing (Simulation Version)

Target Scope: eCommerce / FinTech (Mini Project Implementation)

The Problem: Online transactions often lack trust between buyers and sellers. This mini feature provides a simplified escrow workflow where payments are held (simulated) until delivery is confirmed through a digital signature.

2. Feature Breakdown

Primary Functionality: A simplified escrow system that creates a transaction record, simulates holding buyer payment, allows the seller to mark delivery, captures a digital signature from the buyer as proof of confirmation, and then releases the escrow.

Components Used:

- MongoDB for escrow and signature storage
- Node.js + Express for backend API
- React.js for frontend interface
- react-signature-canvas for digital signature capture

3. Technical Architecture

Data Flow: Buyer creates escrow → Escrow stored (PENDING) → Seller marks delivered → Status (DELIVERED) → Buyer signs digitally → Signature stored → Escrow status updated (RELEASED).

Patterns Used: RESTful API design principles and status-based state transitions.

Modularity: Controllers handle logic, routes manage endpoints, Escrow model defines database structure, and signature capture component is isolated in the frontend.

4. Production Standards Checklist

- Structured error handling for API endpoints
- Idempotent status transitions
- Basic input validation implemented
- README documentation with setup and endpoints

5. Future Scalability

- Real payment gateway integration (e.g., M-Pesa Daraja API)
- JWT authentication and role-based access control
- Automatic escrow expiration
- Dispute resolution workflow
- Admin dashboard
- Email/SMS notifications
- PDF receipt generation with embedded digital signature